

Project

This is the documentation for a project

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1 Introduction

This is the introduction to a project

1.1 Introduction

This is an introduction to the project. This is also the landing page to your book if you chose to publish it. The purpose of this page is to introduce your reader to the project itself, perhaps provide some early documentation of the variables and values of your data, and any idiosyncracies of the analysis itself.

1.2 Notes

This section can be helpful for documenting idiosyncracies of your data, such as:

- Special considerations of the analysis
- Why these analysis were conducted
- When these analyses were conducted

For meta-analysis, a PRISMA statment might be useful here.

1.3 Session

For reproducible results, it might also be helpful to have your ‘sessionInfo()’ documented up front like so:

```
sessionInfo()
```

```
R version 4.3.3 (2024-02-29)
Platform: aarch64-apple-darwin20 (64-bit)
Running under: macOS Sonoma 14.5

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

loaded via a namespace (and not attached):
[1] compiler_4.3.3 fastmap_1.2.0 cli_3.6.2      tools_4.3.3
[5] htmltools_0.5.7 yaml_2.3.8     rmarkdown_2.25 knitr_1.45
[9] jsonlite_1.8.8 xfun_0.41      digest_0.6.34  rlang_1.1.3
[13] evaluate_0.23
```

2 Preparation

Setup materials for the project

2.1 Setup

Here's where I'll call in all my materials necessary for the project.

2.2 Libraries

These materials can include libraries.

```
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.3      v tibble     3.2.1
v purrr      1.0.2      v tidyr      1.3.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

2.3 Functions

These materials can also include custom functions. This is a custom function to round up from .5, which is necessary as the default {round} function rounds to the nearest even digit according to IEC 60559.

These functions can be in blocks below, or as separate R scripts.

```
nround <- function(values, digits) {  
  factor <- 10^digits  
  is_neg <- ifelse(values < 0, -1, 1)  
  result <- is_neg*floor(abs(values)*factor+.5)/factor  
  return(result)  
}
```

3 Data

For importing, cleaning, and manipulating datasets.

3.1 Introduction

Here's where I will import my data, clean it for processing, and manipulate or recode variables to my liking. This might also be a good place to link to the codebook or repeat the contents of the codebook here.

3.2 Data Import

Time to import my data and get to work!

```
source("global.R")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.3      v tibble     3.2.1
v purrr      1.0.2      v tidyr      1.3.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
df_raw <- read.csv("data/validation.csv", header = T)
```

3.3 Data Cleaning

This is a good place to handle missing data or de-identify data.

```
df <- df_raw |>
  select(-id)
```

3.4 Data Manipulation

Every good dataset needs some averages or sums, and we all need to recode a variable or two.

```
df <- df |>
  rowwise() |>
  mutate(job_knowledge_total = sum(jk1,jk2,jk3)) |>
  ungroup() |>
  mutate(job_knowledge_quintile = ntile(job_knowledge_total,5))

df$sex_factor <- df$male |>
  recode(
    `0` = "Female",
    `1` = "Male"
  )
```

3.5 Data Export

People like to have their data in hand. Keeping it in an .Rdata file is great as you can keep the data labelled, but the .csv file is ubiquitous.

```
save(df, file = "data/validation_final.Rdata")
write.csv(df, "data/validation_final.csv")
```

4 Analyses

Conducting statistical analyses to test hypotheses

4.1 Introduction

This is where I will conduct my analyses

4.2 Descriptive Statistics

Perhaps some descriptive statistics would be useful here.

```
source("global.R")
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.3      v tibble     3.2.1
v purrr      1.0.2      v tidyr      1.3.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
load("data/validation_final.Rdata")
```

```
df |>
  group_by(sex_factor) |>
  summarize(mean(job_knowledge_total))
```

```
# A tibble: 2 x 2
  sex_factor `mean(job_knowledge_total)`
  <chr>      <dbl>
1 Female    78.2
2 Male      77.9
```

4.3 Inferential Statistics

Some simple statistics are presented below to evaluate the data.

```
t.test(job_knowledge_total ~ sex_factor, data = df)
```

Welch Two Sample t-test

```
data: job_knowledge_total by sex_factor
t = 0.44475, df = 186.49, p-value = 0.657
alternative hypothesis: true difference in means between group Female and group Male is not 0
95 percent confidence interval:
 -1.124555  1.779189
sample estimates:
mean in group Female    mean in group Male
      78.23853          77.91121
```

5 Figures

Time to make some data visualizations

5.1 Setup

As with the data and analysis sections, we'll need to call in our global scripts and the final dataset made earlier.

```
source("global.R")
load("data/validation_final.Rdata")
```

5.2 Data Visualization

Here's a good spot to make some data visualizations for exploratory or event publication ready figures. Adding this reference to Figure 5.1 allows readers to hover over the refence to preview the figure which can be helpful if you have long blocks of text and are discussing a figure which may not appear for a while.

<https://github.com/quarto-dev/quarto-cli/discussions/4655>

If you're just going to have an HTML document without a PDF, I'd recommend leaning on the `embed-resources: true` in the `_quarto.yml` file. When this is used, you don't need to seperately store your figures and then display them using Markdown (e.g., `![]()`). They can, inseed, simply be labelled in the R code chunk itself (e.g., `#| label: fig-1`)

5.3 Simple Figure

Figure 5.1 is an example of a simple figure which can quickly be put together for exploratory purposes. You'll notice the Figure 5.1 is automatically named when the document is numbered with the option `numbered-sections: true` in the `_quarto.yml` file. This is also the default. If you chose `false` the figure and cross-reference will be titled whatever is in the parenthesis.

```
fig1 <- ggplot(data = df, aes(x = lapr, y = factor(job_knowledge_quintile))) +  
  geom_bar(stat = "summary", fun = mean, width = .5)
```

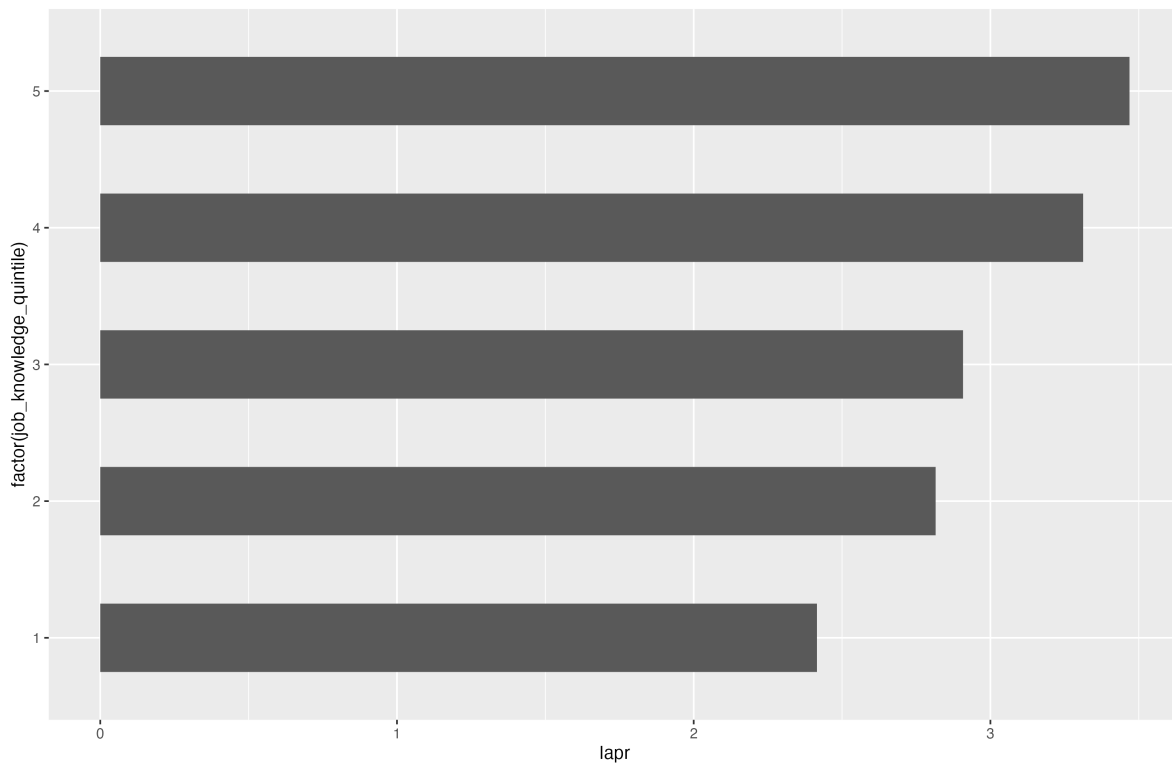


Figure 5.1

5.4 Publication Ready Figure

Figure 5.2 is a the same figure as above, just made a little more polished for publication.

```
fig2 <- ggplot(data = df, aes(x = lapr, y = factor(job_knowledge_quintile))) +
  geom_bar(stat = "summary", fun = mean, width = .5) +
  geom_text(aes(label = nround(after_stat(x),2)),
            stat = "summary",
            fun = "mean",
            size = 20/.pt,
            hjust = -.2) +
  coord_cartesian(xlim = c(1,5)) +
  labs(
    title = "Average Performance by Job Knowledge Quintile",
    x = "Average Last Annual Performance Review Rating",
    y = "Job Knowledge Test Quintile"
  ) +
  theme_minimal(base_size = 20) +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 20),
    panel.grid = element_blank()
  )
```

5.5 Figure Export

It's often important to have figures in a format suitable for manuscripts or other programs. We also need to export the figures for later reference in markdown format using `![]()`.

```
ggsave(
  plot = fig1,
  filename = "figures/Figure1.png",
  height = 6.5,
  width = 10,
  units = "in"
)

ggsave(
  plot = fig2,
  filename = "figures/Figure2.png",
  height = 6.5,
  width = 10,
  units = "in"
)
```

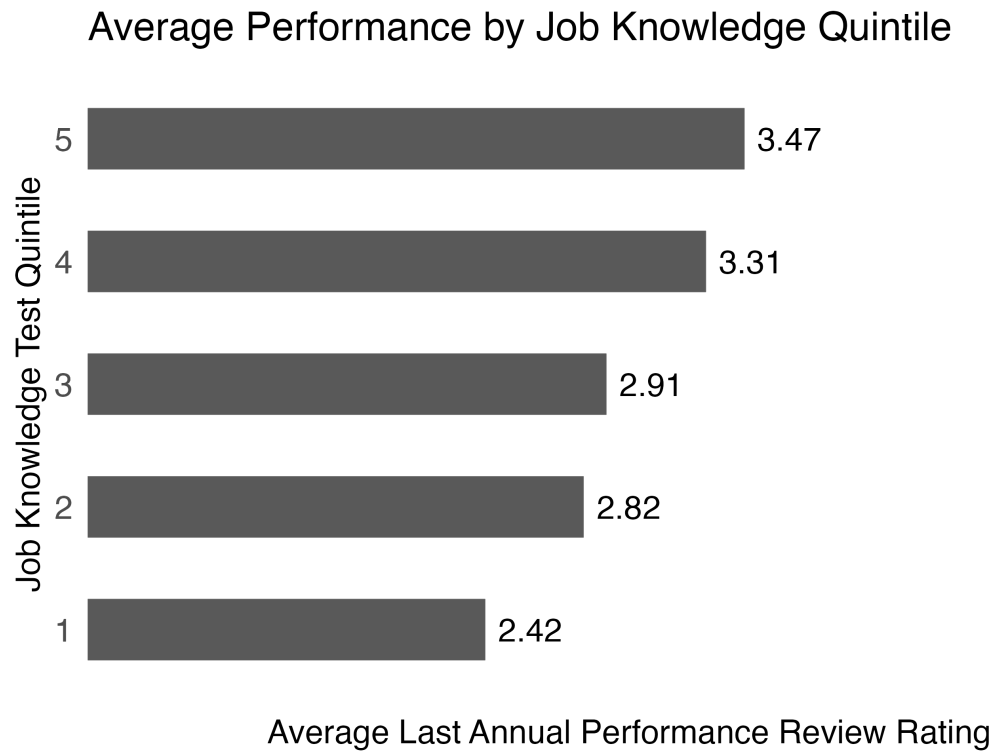


Figure 5.2